
ME 617: Mechanics of Composite Materials

Course Code	ME 617
Course Name	Mechanics of Composite Materials
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solids
Intended for	MS/PhD
Distribution	Elective
One-time approval	14th BoA (Sep 2016); 12th Senate (Mar 2017)

Course Contents

- **Module I**

Introduction, Classification of Composite Materials, advantage of composite materials and applications [4 Lectures]

- **Module II**

Macro-mechanical behaviour of a lamina; Stress strain relationships for anisotropic materials, stiffness's and compliances, restrictions of engineering constants, invariant properties of lamina, Strength of lamina [10 Lectures]

- **Module III**

Micromechanical behaviour of lamina: Mechanics of materials approach for stiffness, elasticity approach, imperial approaches, strength estimations [12 Lectures]

- **Module IV**

Micromechanical behaviour of Laminates: Classical laminate theory, special cases of laminates, strength of laminates, Interlaminar stresses [12 Lectures]

- **Module V**

Experimental characterization of composites: Uniaxial, biaxial tension and compression tests, interfacial and inter laminate shear stress characterizations. [4 Lectures]

Text and Reference Books

1. Laszolo P Kollar and George S. Springer, **Mechanics of Composite Structures**, Cambridge university Press, 2003.
2. J. N. Reddy, **Mechanics of Laminated Composite Plates and Shells Theory and Analysis**